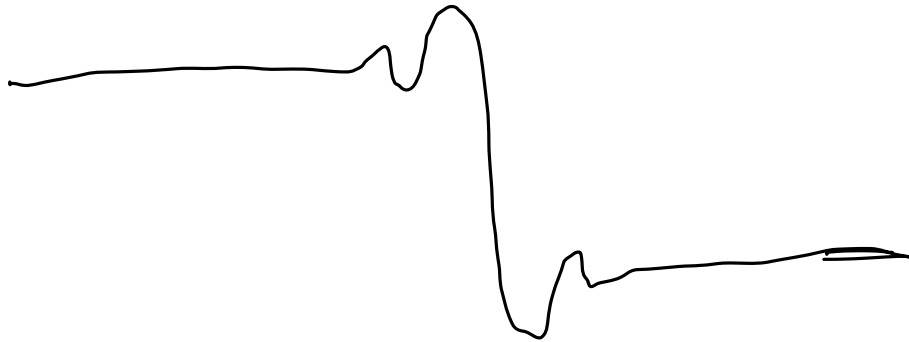
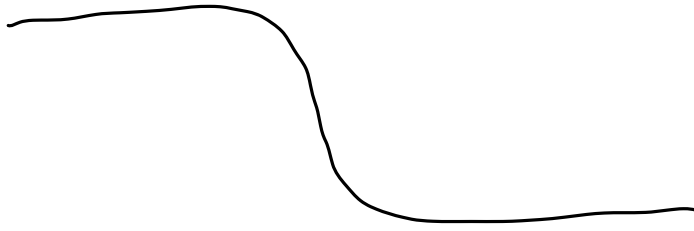


$$U_j'(t) = - \frac{U_{j+1} - U_{j-1}}{2\Delta x}$$

$$u_t = -u_x$$



# PDEs from a spectral perspective

$$u_t + au_x = 0 \quad -\pi < x \leq \pi$$

$$u(x, t=0) = u_0(x)$$

$$u(-\pi, t) = u(\pi, t)$$

Fourier ansatz:  $u(x, t) = e^{i(kx - \omega t)}$

$$u_t = -i\omega u$$

$$\Rightarrow -i\omega u = -aik u \Rightarrow \omega(k) = ak$$

$$u_x = ik u$$

Dispersion relation

So we have

$$u(x, t) = e^{i(kx - akt)} = e^{ik(x - at)}$$

Speed:  $\frac{\omega(k)}{k} = a$

## Dispersion

$$u_t + au_x + \varepsilon u_{xxx} = 0$$

$$\partial_t \rightarrow i\omega \quad \partial_x \rightarrow ik$$

$$-i\omega + iak - i\varepsilon k^3 = 0$$

$$\omega - ak + \varepsilon k^3 = 0$$

$$\omega(k) = ak + \varepsilon k^3$$

$$u(x, t) = e^{ik(x - (a + \varepsilon k^2)t)}$$

$$c = \frac{\omega(k)}{k} = a + \varepsilon k^2$$

Wavenumber-dependent

Notice that  $c \rightarrow \infty$  as  $|k| \rightarrow \infty$ .

$$u_t + au_x = \varepsilon u_{xx} \quad (ik)^2 (-i\omega)$$

$$-i\omega + aik = i\varepsilon k^2 \omega$$

$$\omega - ak = -\varepsilon k^2 \omega$$

$$(1 + \varepsilon k^2) \omega = ak$$

$$\omega = \frac{ak}{1 + \varepsilon k^2} \quad c = \frac{\omega}{k} = \frac{a}{1 + \varepsilon k^2}$$

$\varepsilon > 0$ :  $|c|$  is bounded for all  $k$

## Dissipation

$$u_t = au_{xx}$$

$$u(x,t) = e^{ikx} e^{-ak^2 t}$$

$$-i\omega = -ak^2$$

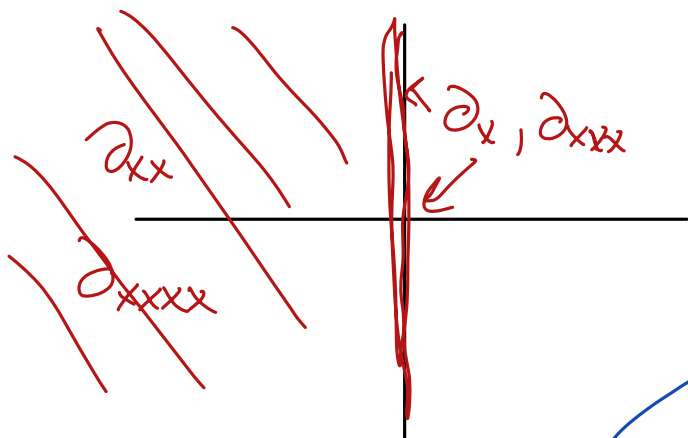
$$\omega = -iak^2$$

## General-order<sup>(space)</sup> derivatives

$$u_t = \sum_{j=0}^m a_j \partial_x^j u$$

$$-i\omega = \sum_{j=0}^m a_j (ik)^j = \underbrace{\sum_{j \text{ even}} \pm a_j k^j}_{\text{growth or decay}} \pm$$

$$i \underbrace{\sum_{j \text{ odd}} a_j k^j}_{\text{oscillation (dispersion)}}$$



$$u_{tt} = v_{xt}$$

$$v_{xt} = v_{tx}$$

$$v_{tx} = u_{xx}$$

Systems

$$\left. \begin{array}{l} u_t = v_x \\ v_t = u_x \end{array} \right\}$$

$$\Leftrightarrow u_{tt} = u_{xx}$$

$$-\omega^2 = -k^2$$

$$\omega = \pm |k| \Rightarrow c = \pm 1$$

$$\cancel{i\omega u} + \cancel{ikv} = 0$$

$$\cancel{i\omega v} + \cancel{iku} = 0$$

$$\begin{bmatrix} \omega & k \\ k & \omega \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 0$$

$$\det \begin{bmatrix} \omega & k \\ k & \omega \end{bmatrix} = \omega^2 - k^2 = 0$$

$$\frac{\vec{v}}{t} + A \partial_x^j \vec{v} = 0$$

$$A \text{ diagonalizable: } A = R \Lambda R^{-1}$$

$$\Lambda: \text{diagonal} \quad \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

$$\vec{x} = R^{-1} \vec{v}$$

$$-i\omega \vec{v} + (ik)^j R \Lambda R^{-1} \vec{v} = 0$$

$$-i\omega R^{-1} \vec{v} + (ik)^j \Lambda R^{-1} \vec{v} = 0$$

$$-i\omega \vec{x} + (ik)^j \Lambda \vec{x} = 0$$

$$-i\omega + (ik)^j \lambda_p = 0 \Rightarrow -i\omega = -(ik)^j \lambda_p$$

For example:

$$u_t = v_{xx}$$

$$v_t = -u_{xx}$$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \lambda = \pm i$$

This is dispersive.

In fact

$$\frac{d}{dt} \int (u^2 + v^2) dx = 0.$$

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## FD discretizations

$$u_t + au_x = 0$$

Spatial discretization:

Upwind:  $u_x|_{x=x_j} \approx \frac{U_j - U_{j-1}}{\Delta x}$

Centered:  $u_x|_{x=x_j} \approx \frac{U_{j+1} - U_{j-1}}{2\Delta x}$

Semi-discrete dispersion analysis

$$U_j(t) \rightarrow g(t)e^{ikx_j} = g(t)e^{ikj\Delta x}$$

Upwind  $U'_j(t) = -a \frac{U_j - U_{j-1}}{\Delta x} \quad \leftarrow U'(t) = A U(t)$

$$g'(t)e^{ikj\Delta x} = -ag(t) \frac{e^{ikj\Delta x} - e^{ik(j-1)\Delta x}}{\Delta x}$$

$$g'(t) = \frac{-ag(t)}{\Delta x} (1 - e^{-ik\Delta x})$$

$$g'(t) = \lambda g$$

$$\lambda = \frac{-a}{\Delta x} (1 - e^{-ik\Delta x})$$

$$\lambda = \frac{-a}{\Delta x} \left( 1 - \left( 1 - ik\Delta x + \frac{(ik\Delta x)^2}{2} + O((k\Delta x)^3) \right) \right)$$

$$g(t) = e^{\lambda t} g(0)$$

$$= -a \left( ik + \frac{k^2 \Delta x}{2} \right) + O(k^3 \Delta x^2)$$

$$= -aik - a \frac{k^2 \Delta x}{2}$$

← numerical dispersion

So

$$U_j(t) \approx \exp \left[ \left( -aik - a \frac{k^2 \Delta x}{2} \right) t \right] e^{ikj\Delta x} \quad (1)$$

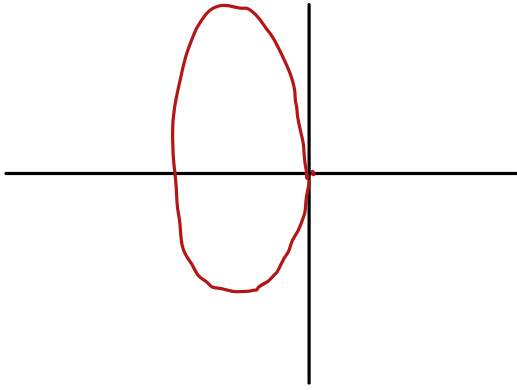
← numerical dissipation

Recall the disp. relation for the PDE:

$$u(x,t) = e^{ikx} e^{-ikat} \quad (2)$$

Comparing (1) and (2) we see that the semi-discretization adds artificial dispersion and dissipation.

The eigenvalues we've found lie on an ellipse in the complex plane:



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Homework: Find the dispersion relation for the centered FD discretization of the advection equation.